

# Technical Overview

## Smart i-LAB Testbed

Frontend, backend, architecture, CI/CD,  
ML training, and operations

[pctiope/smart-i-lab-testbed](https://github.com/pctiope/smart-i-lab-testbed)

May 2026

# Objective

## Review question

How does the combined Smart i-LAB repository move from lab devices to API, frontends, model training, deployment, and operational verification?

Repo-grounded

No runtime secrets

No generated model data

- Keep the Metropolis Beamer base, but use a darker presentation rhythm.
- Show flows and ownership boundaries instead of dense path tables.
- Separate IoT1 cutover status from Zone 5 CI/CD and ML delivery.
- Describe Zone 5 and AIR1 model contracts without inventing metrics.



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**Frontend**



Tracked Digital Twin floor asset.

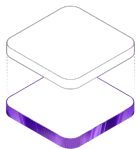
## Purpose

- Web visualization for Room 308 device status and selected controls.
- Demonstrates the IoT1 REST API as the user-facing integration layer.
- Served as static assets by nginx after a Vite build.

## Boundary

- API base is configured at build time with `VITE_API_URL`.
- Three.js assets stay with the tracked frontend source.
- It does not own model inference or data collection.

# Zone 5 Dashboard



Tracked dashboard visual reference.



Desk 5 ROI mask used by the CV counter.

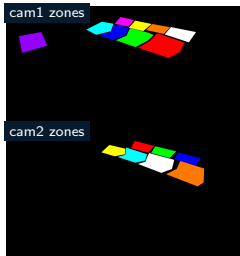
## What the operator sees

- Occupancy probability, CV truth, recent history, squared error, and sensor context.
- Annotated MJPEG from the file-backed latest frame path.
- Split-pane layout with the camera feed beside the readout surface.

## Backend contract

`/api/health`, `/api/current`, `/api/history`, `/api/stream`, and `/api/video.mjpg` are preserved so UI changes do not redefine runtime APIs.

# AIR1 All-Zones Dashboard



Tracked AIR1 zone masks define the dashboard's per-zone camera coverage.

## Operational view

- One shared AIR1 model reports per-zone probabilities for zones 1–15.
- Two camera streams feed separate zone masks and one per-zone label table.
- The app can remain up while model readiness is blocked by a missing production pointer.

## Important separation

- AIR1 excludes power and mmWave from model inputs.
- Missing camera coverage remains null; it is not converted to vacant.



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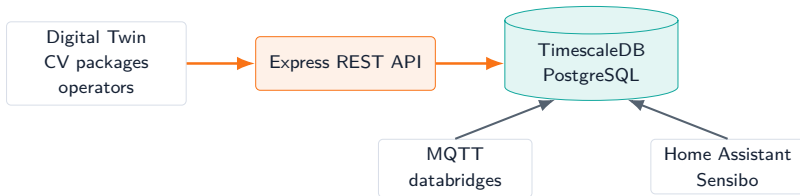
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## Backend

# IoT1 REST API

## Role

- Express API for sensor reads, device registries, transactions, groups, logs, and actuator paths.
- Digital Twin and CV packages consume the API instead of talking directly to device tables.
- Device histories are stored in PostgreSQL with TimescaleDB hypertables.





# Security And Storage

## Auth

- API-key checks.
- Access-level validation.
- CV clients continue to send x-api-key.

## Hardening

- Parameterized SQL or allow-lists.
- Helmet, rate limits, body caps.
- Non-root containers.

## Storage

- Versioned migrations.
- Sensor tables created lazily.
- Dirty legacy tables remain cleanup work.

`docs/IOT1_AUDIT.md`

`docs/IOT1_REMEDIATION.md`

`migrations/apply.py`

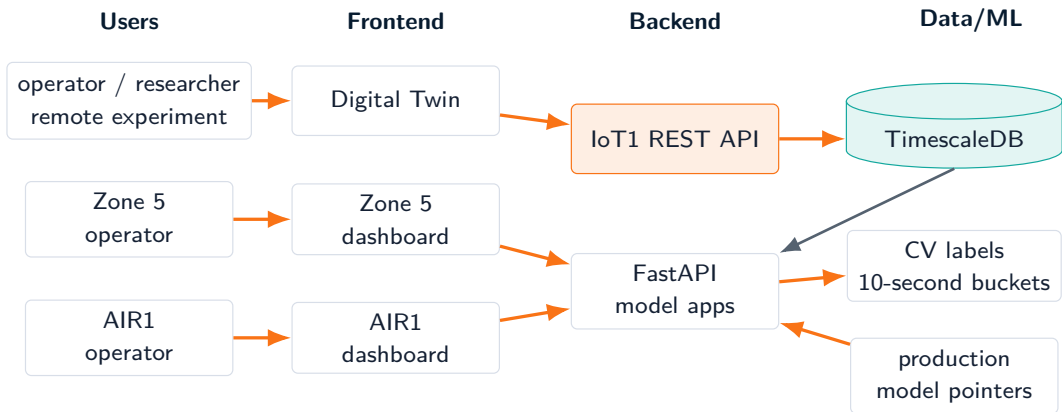


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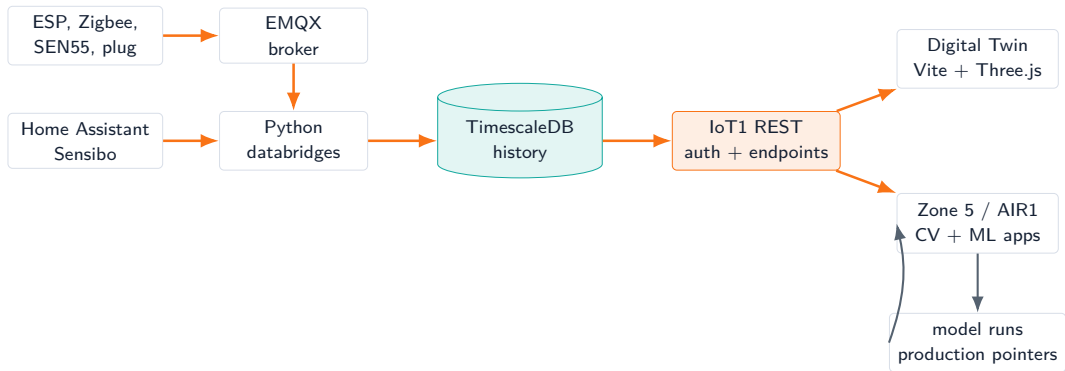
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# Architecture

# User And Data Flow



# End-To-End Architecture



Device data

REST integration

CV labels

Model delivery

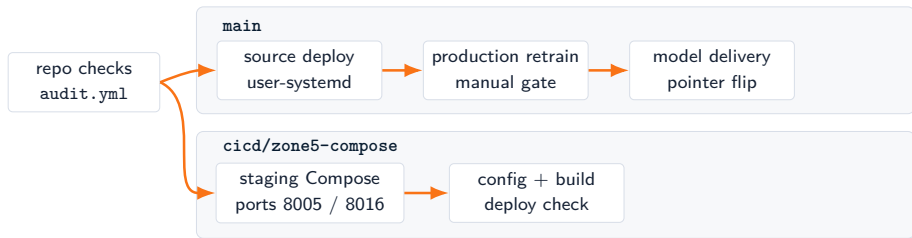


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CI/CD

# GitHub Workflow Topology



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## Workflow

## Ownership

|                                 |                                                                      |
|---------------------------------|----------------------------------------------------------------------|
| zone5-systemd-source-deploy.yml | Production source deploy; restarts live services only.               |
| zone5-compose-ci-cd.yml         | Staging Compose path; self-hosted deploy to separate ports.          |
| zone5-production-retrain.yml    | Starts the trainer and packages a promoted model only after gating.  |
| zone5-model-delivery.yml        | Installs a vetted model artifact and updates the production pointer. |

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# Staging Compose Vs. Production systemd

## Production

- Branch: `main`
- Runtime: user-level `systemd`
- Ports: backend 8000, Vite 8015
- Source deploy refuses active trainer collisions.

## Staging

- Branch: `cicd/zone5-compose`
- Runtime: Docker Compose
- Ports: backend 8005, Vite 8016
- Copies env and model pointer from production when needed.

**Source deploy, staging deploy, training, and model delivery stay separate.**

# Retrain And Model Delivery



- Scheduled retraining is present but disabled until the manual proof run is accepted.
- Promotion failure leaves collection and serving alone.
- Sanitized model registry metadata can be committed; generated run directories stay off-source.





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## ML Training

# CV Labels

## Zone 5

- One camera, one desk mask, one 10-second Zone 5 label stream.
- Target is `zone_occupied`.
- Occupied when median person count crosses the configured threshold.

## AIR1 all-zones

- Two cameras, per-zone masks, long-form labels keyed by `timestamp + zone_id`.
- Target is `occupied`.
- Missing camera coverage remains null instead of becoming a negative label.



# Zone 5 Methodology

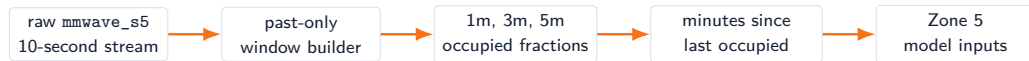
## Feature families

- AIR-1 Zone 5: temperature, humidity, CO2, PM2.5.
- Smart plug: `power_s5`.
- mmWave: `mmwave_s5` and recency context.
- SEN55: particulate, environment, VOC, NO<sub>x</sub>.
- Time: hour and day-of-week cycles.

## Training contract

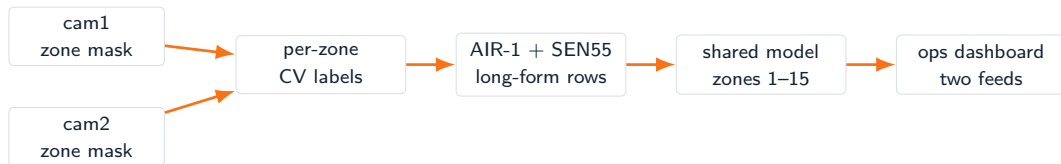
- Current marker:  
`zone5_mmwave_recency_v1`.
- Missing indicators are diagnostics and metadata, not model inputs.
- Legacy  
`zone5_missingness_decoupled_v1`  
artifacts remain loadable for live compatibility.
- Trainer snapshots the rolling CSV before fitting.

# mmWave Recency Contract



- mmWave remains an input feature, not the training target.
- Recency features are computed from prior samples only.
- The model sees short-history motion context without future leakage.
- Dashboard history surfaces the lookback context beside probability.

# AIR1 All-Zones Pipeline



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## Contract guard

zone\_id is retained but not an input  
Power and mmWave are excluded  
Production pointer gates inference

## Why it matters

Supports grouping, splits, audit, and display only.  
AIR1 parity does not import Zone 5-specific signals.  
The UI can be healthy before a promoted model exists.

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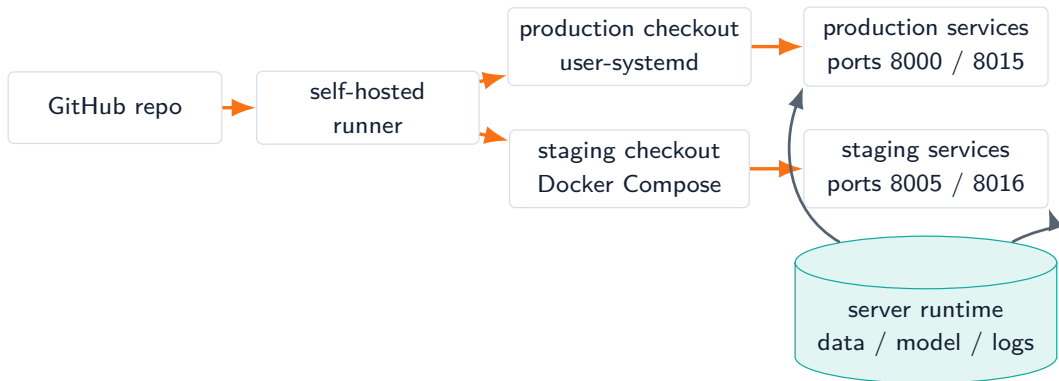


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## Deployment

# Deployment Architecture



Runtime credentials, generated data, model runs, and logs live on the server, not in the presentation or source history.

# Verification And Performance Gates

## CI gates

- Python compile checks for package and app modules.
- Zone 5 ground-truth contract tests.
- Vite build checks.
- Systemd unit verification.
- Docker Compose config and image build.

## Runtime gates

- `/api/health` confirms app health and model readiness fields.
- Model promotion checks smoke loading and non-regression conditions.
- IoT1 cutover still requires backup, migrations, deploy, smoke test, and log tail.
- Latency and label-balance monitoring remain roadmap items.



# Roadmap And Risks

## Near-term roadmap

1. Complete the IoT1 production cutover sequence.
2. Dedupe legacy per-device timestamp histories.
3. Prove manual Zone 5 retraining before enabling scheduled retrain.
4. Add model freshness, promotion, and upstream API latency monitoring.

## Risks to keep visible

- Live QA depends on lab network, TimescaleDB, EMQX, Home Assistant, cameras, and devices.
- Training quality depends on both-class CV labels across usable calendar windows.
- Contract changes require artifact compatibility and dashboard context updates.

# Source Anchors

| Topic                            | Primary source paths                                                                           |
|----------------------------------|------------------------------------------------------------------------------------------------|
| IoT1 status and cutover          | docs/STATUS.md, docs/IOT1_PRE_DEPLOYMENT.md,<br>docs/IOT1_REMEDIATION.md                       |
| Digital Twin                     | Smart-iLab_DigitalTwin/, README.md                                                             |
| Zone 5 ML and operations         | zone5_cv_time_features_package/PIPELINE.md,<br>zone5/.../feature_contract.py                   |
| AIR1 all-zones ML and operations | air1_all_zones_cv_time_features_package/PIPELINE.md,<br>air1_all_zones/.../feature_contract.py |
| CI/CD and deployment             | .github/workflows/, zone5/.../CICD_SYSTEMD.md,<br>zone5/.../CICD_DOCKER_COMPOSE.md             |

**Closing point:** source, staging runtime, production runtime, and model artifacts each have a separate owner and gate.